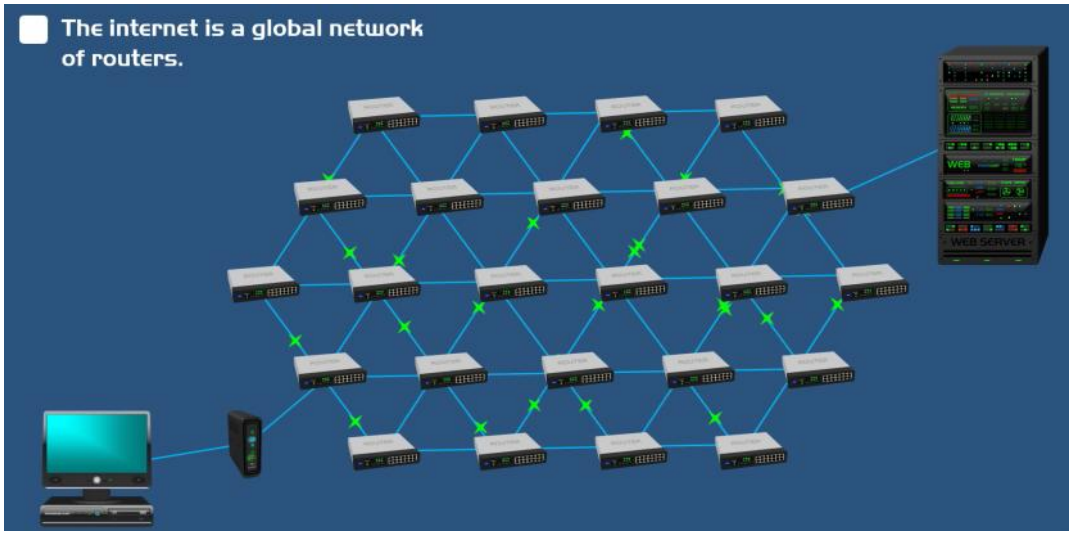
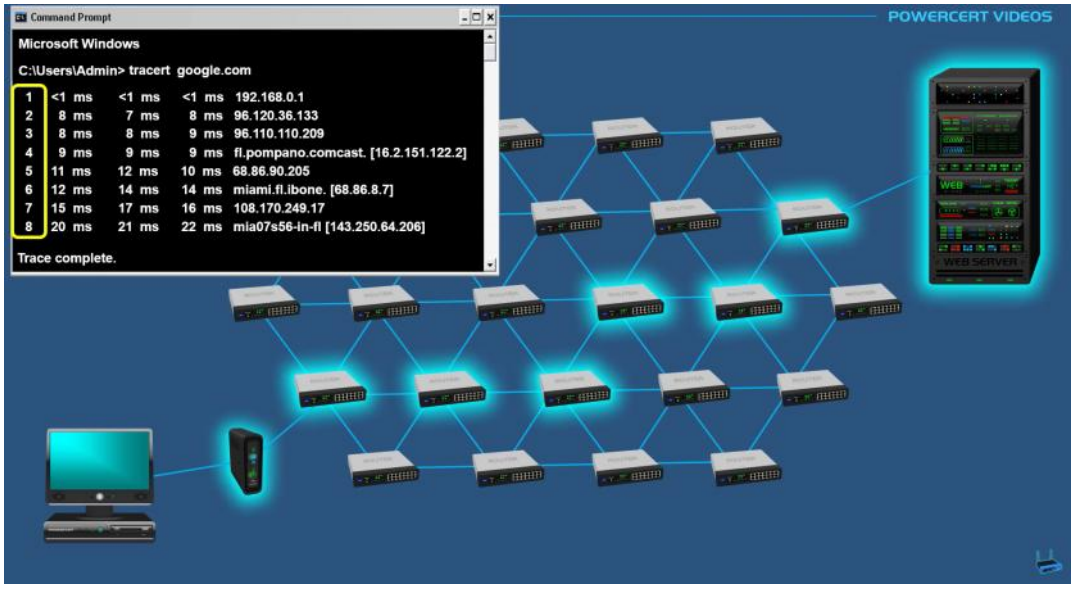
Traceroute (tracert)

Saturday, March 14, 2026 8:31 PM

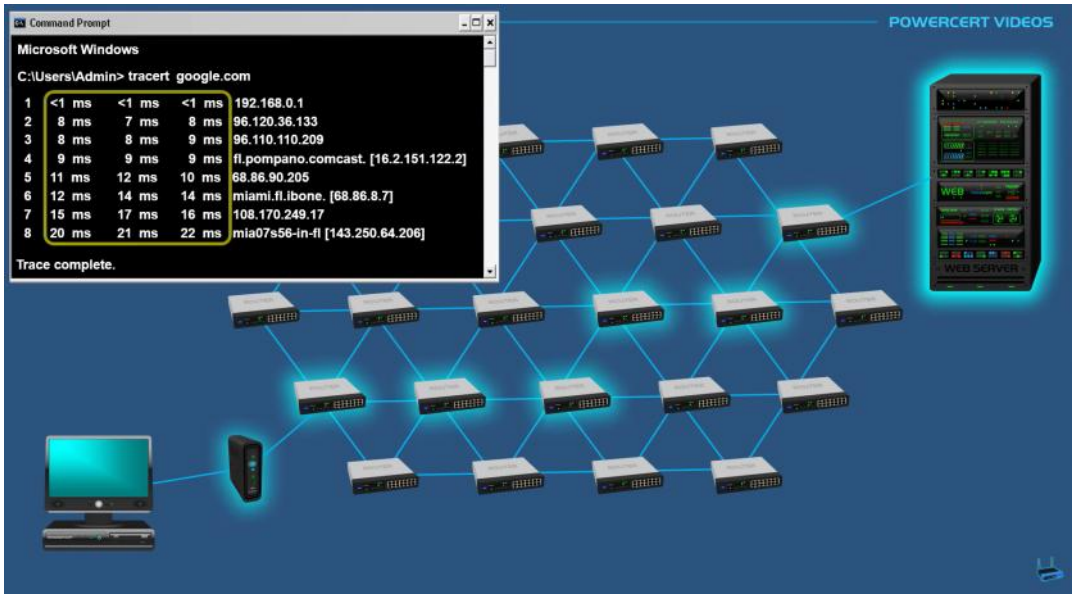
Traceroute is a command line network diagnostic tool used to discover the path that data packets take from a source computerto a destination on the internet. The internet is made up of many routers, and traceroute shows each router called a hop that packets pass through on the way to the destination.



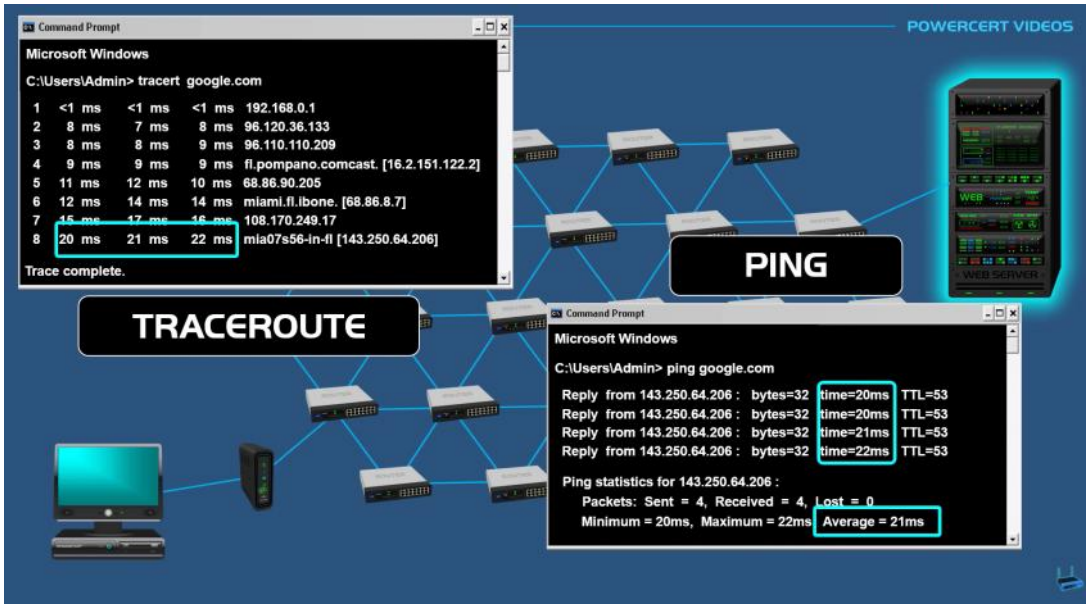
Traceroute works by sending packets with increasing TTL Time To Live values. Each router that receives the packet decreases the TTL by one. When TTL reaches zero the router sends a reply back showing its IP address and the round trip time RTT. This process continues until the packets reach the final destination.



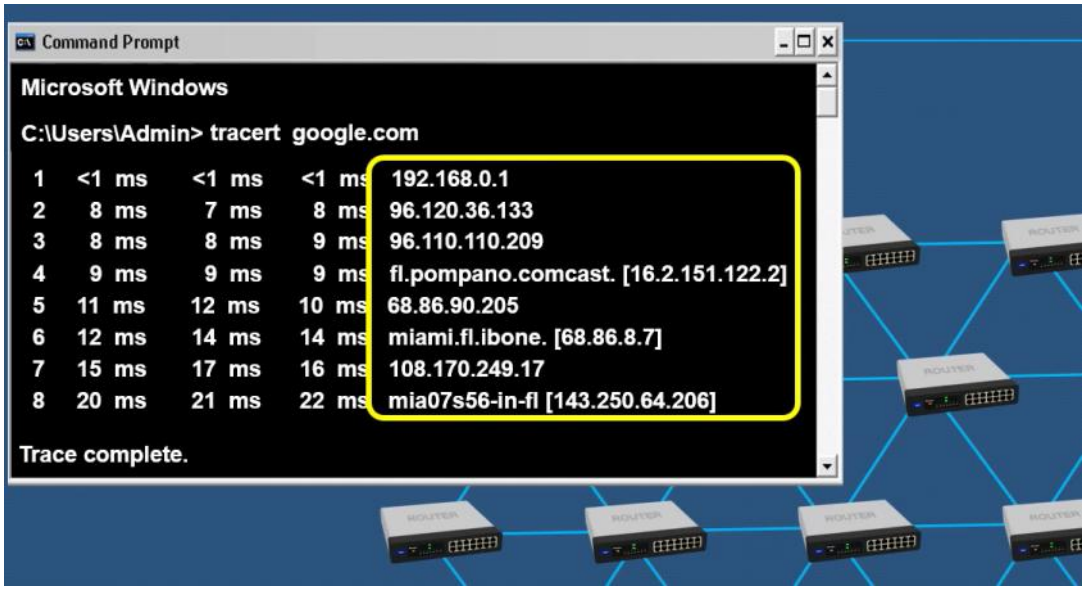
Number of hop/steps it took to the destination



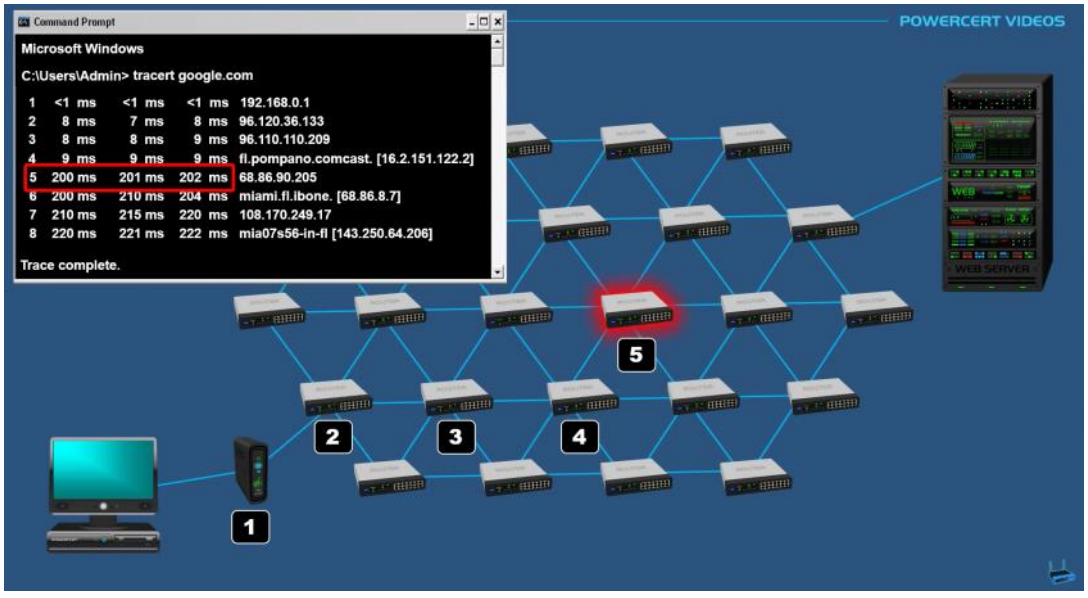
Time taken for data packet to take round trip from the router to the source



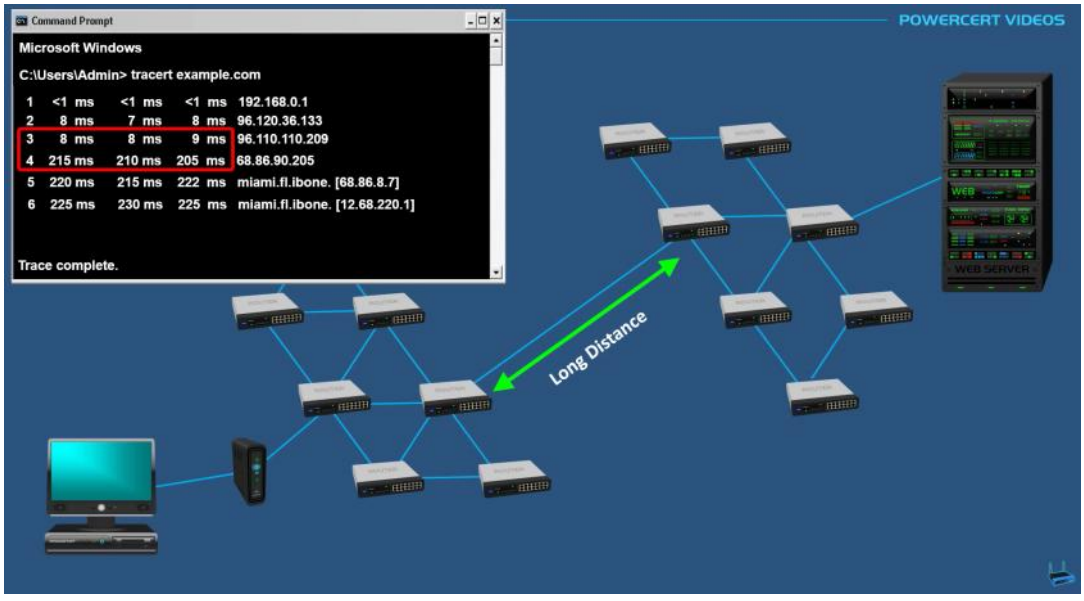
Ping and traceroute would take almost the same time



This shows hostname (if available) or IP addresses of the router/destination



Sometimes there can be some issues with router as it may take too much time to hop also there can be other issues such as thedistance between the routers as shown below



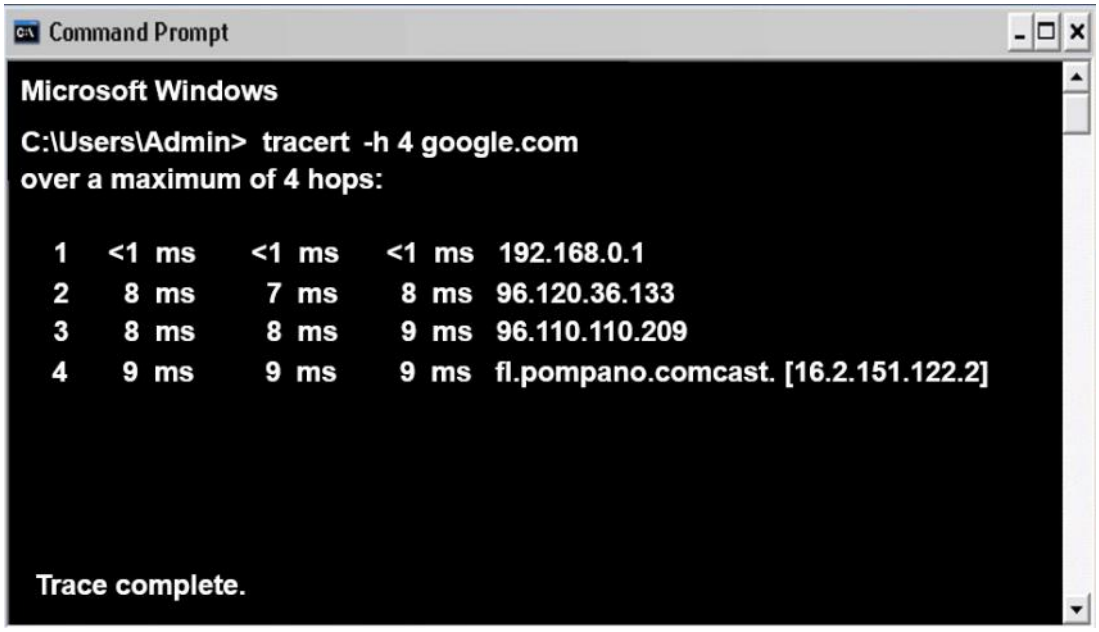
For every hop traceroute usually sends three packets and displays three RTT values in milliseconds. This helps measure the delay between the source and each router along the path.

```
Command Prompt
Microsoft Windows
C:\Users\Admin> tracert google.com
  1   <1 ms    <1 ms    <1 ms  192.168.0.1
  2    8 ms     7 ms     8 ms  96.120.36.133
  3    8 ms     8 ms     9 ms  96.110.110.209
  4    *        *        *     Request timed out.
  5   11 ms    12 ms    10 ms  68.86.90.205
  6   12 ms    14 ms    14 ms  miami.fl.ibone. [68.86.8.7]
  7   15 ms    17 ms    16 ms  108.170.249.17
  8   20 ms    21 ms    22 ms  mia07s56-in-fl [143.250.64.206]
Trace complete.
```

This can mean that there is some issue with the router or it can also mean that the router is not set to give the traceroute response but still it passed the request to the next router

```
Command Prompt
Microsoft Windows
C:\Users\Admin> tracert google.com
over a maximum of 30 hops:
  1   <1 ms    <1 ms    <1 ms  192.168.0.1
  2    8 ms     7 ms     8 ms  96.120.36.133
  3    8 ms     8 ms     9 ms  96.110.110.209
  4    9 ms     9 ms     9 ms  fl.pompano.comcast. [16.2.151.122.2]
  5   11 ms    12 ms    10 ms  68.86.90.205
  6   12 ms    14 ms    14 ms  miami.fl.ibone. [68.86.8.7]
  7   15 ms    17 ms    16 ms  108.170.249.17
  8   20 ms    21 ms    22 ms  mia07s56-in-fl [143.250.64.206]
Trace complete.
```

After 30 Hops, the data packet will be dropped as the TTL(Time to Live) is only 30hops limited



We can set the custom TTL for testing as the data packet will drop after the custom set hop number

Traceroute is different from ping because ping only checks connectivity with the final destination and measures overall RTT. Traceroute shows every intermediate router which helps identify where delays or network problems occur.

In traceroute output a gradual increase in RTT is normal as packets travel farther. Sudden high delays may indicate congestion or long distance links. Asterisks mean the router did not send a reply but it may still be forwarding packets.

Traceroute is commonly used by network engineers to troubleshoot connectivity issues, locate bottlenecks and understand the routing path between networks.

TTL prevents data packets from traveling endlessly around the internet especially if the router is misconfigured then the data may travel endlessly causing high bandwidth

In Linux /mac its - traceroute example.com while in windows its tracert example.com